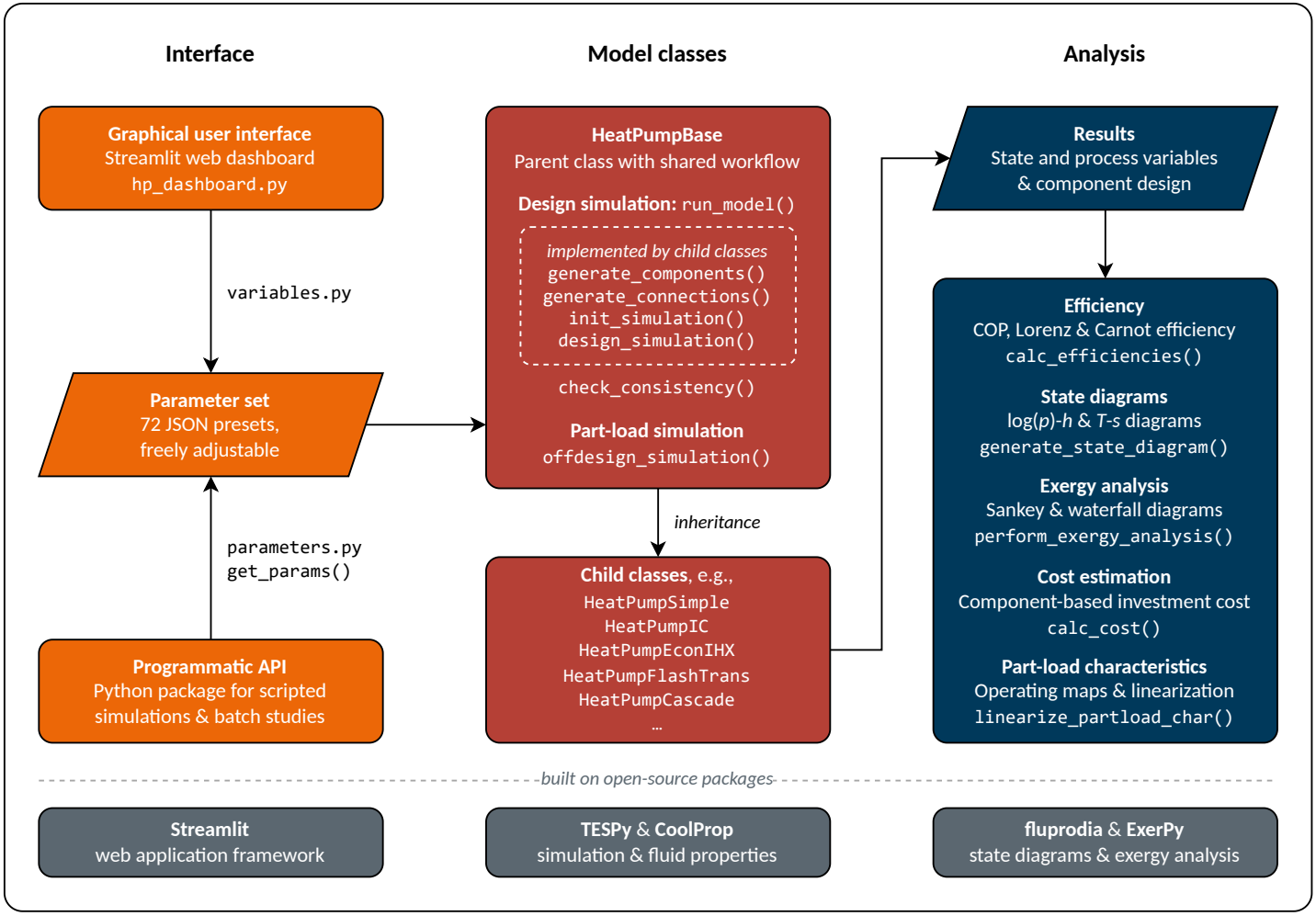

Interface
Model classes
Analysis
Graphical user interface
Streamlit web dashboard
hp_dashboard.py
variables.py
Parameter set
72 JSON presets,
freely adjustable
parameters.py
get_params()
Programmatic API
Python package for scripted
simulations & batch studies
HeatPumpBase
Parent class with shared workflow
Design simulation: run_model()
implemented by child classes
generate_components()
generate_connections()
init_simulation()
design_simulation()
check_consistency()
Part-load simulation
offdesign_simulation()
inheritance
Child classes, e.g.,
HeatPumpSimple
HeatPumpIC
HeatPumpEconIHX
HeatPumpFlashTrans
HeatPumpCascade
...
Results
State and process variables
& component design
Efficiency
COP, Lorenz & Carnot efficiency
calc_efficiencies()
State diagrams
log(p)-h & T-s diagrams
generate_state_diagram()
Exergy analysis
Sankey & waterfall diagrams
perform_exergy_analysis()
Cost estimation
Component-based investment cost
calc_cost()
Part-load characteristics
Operating maps & linearization
linearize_partload_char()
built on open-source packages
Streamlit
web application framework
TESPy & CoolProp
simulation & fluid properties
fluprodia & ExerPy
state diagrams & exergy analysis